

RESEARCH ARTICLE

The dynamics of immune response in COVID-19 patients with different illness severity

Bo Zhang | Daoyuan Yue | Yun Wang | Feng Wang | Shiji Wu  | Hongyan Hou

Department of Laboratory Medicine, Tongji Hospital, Tongji Medical College, Huazhong University of Science and Technology, Wuhan, China

Correspondence

Shiji Wu and Hongyan Hou, Department of Laboratory Medicine, Tongji Hospital, Tongji Medical College, Huazhong University of Science and Technology, 1095 Jiefang Rd, 430030, Wuhan, China.

Email: wilson547@163.com and houghongyan89@163.com

Funding information

National Mega Project on Major Infectious Disease Prevention, Grant/Award Number: 2017ZX10103005-007

Abstract

This study aimed to analyze the dynamic changes of lymphocyte subsets and specific antibodies in coronavirus disease 2019 (COVID-19) patients with different illness severity. The amounts of lymphocyte subsets and the levels of immunoglobulin M (IgM) and IgG antibody were retrospectively analyzed in 707 COVID-19 cases. The amounts of lymphocyte subsets were significantly decreased with the increased severity of illness and the levels of IgM and IgG were lower in critical cases than severe and moderate cases. In deceased patients, the lymphocytes subsets were significantly lower than recovered patients. However, the relationship between the levels of IgM and IgG and the amounts of lymphocyte subsets were not significantly correlated. During different stages of COVID-19, the total T cell, CD4⁺ T cell, and CD8⁺ T cell counts were gradually recovered to the normal levels in severe and critical groups but the changing trend was relatively stable in the moderate group. The production of IgM and IgG antibodies were delayed in critical groups but also could reach the peak levels at one month after illness onset and decreased to background levels. To detect the kinetics of lymphocytes and antibodies has important clinical value in predicting the illness severity and understanding the pathogenesis of COVID-19.

KEYWORDS

antibody, COVID-19, illness severity, lymphocyte subsets, SARS-CoV-2

1 | INTRODUCTION

The coronavirus disease 2019 (COVID-19) is an infectious disease caused by severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) and has spread quickly worldwide.^{1–3} The outbreak of COVID-19 has caused considerable effects on the global public health community and economy.^{4,5} During SARS-CoV-2 infection, host factors trigger an immune response against the virus. Immune insufficiency or misdirection may increase viral replication and cause

tissue damages whereas overactive immune responses may induce immunopathological conditions.⁶ Host immune responses play a vital role during the clearance of viral infection. Therefore, evaluating the kinetics of the cellular and humoral immune responses in COVID-19 patients with different severity has potential significance.

Lymphocytes and the subsets of CD4⁺ T cells, CD8⁺ T cells, B cells, and natural killer (NK) cells play an important role in the maintenance of immune system function. The alteration of lymphocyte subsets varied from the virus types and might indicate a potential association with the viral mechanism. T cells, including CD4⁺ T cells and CD8⁺ T cells, played an important antiviral role in balancing the combat against pathogens and the risk of developing autoimmunity or

Abbreviations: ARDS, acute respiratory distress syndrome; COVID-19, coronavirus disease 2019; IQR, interquartile range; NK, natural killer; SARS-CoV-2, severe acute respiratory syndrome coronavirus 2; SD, standard deviation.

overwhelming inflammation.⁷ The decrease of CD8⁺ T cells and B cells and the increase of CD4/CD8 ratio have been reported in severe COVID-19 cases.⁸ Moreover, the decrease of lymphocyte not only correlated with the illness severity of COVID-19 but also an important risk factor for the poor prognosis of SARS-CoV-2 infection.^{9,10}

It has been widely accepted that immunoglobulin M (IgM) provides the first line of defense during viral infections, before the generation of adaptive, high-affinity IgG responses that are important for long-term immunity and immunological memory.¹¹ The specific antibody response against SARS-CoV-2 has been reported to be correlated with the illness severity and prognosis of COVID-19.^{12,13} Therefore, it is particularly important to observe the relationship between antibody response and cellular immunity and the dynamics of these marker changes in COVID-19 patients with different illness severity. However, there are few studies to evaluate the dynamic changes of antibody levels and lymphocyte subsets comprehensively in COVID-19 patients with different disease severity.

In this study, we longitudinally characterized the changes of antibodies and lymphocyte subsets in the peripheral blood of COVID-19 patients with different illness severity, which has potential value for further understanding the immunopathological mechanism during SRAS-COV-2 infection.

2 | METHODS

2.1 | Patients

This study retrospectively collected COVID-19 patients who have detected lymphocyte subsets and SARS-CoV-2 antibodies in Tongji Hospital, Wuhan, China during the same day between February and April, 2020. The confirmed diagnosis of COVID-19 was defined as a positive result using real-time reverse transcription-polymerase chain reaction detection from routine nasal and pharyngeal swab specimens. A total of 707 patients were included and the demographic, clinical information, laboratory results were finally collected from electronic medical records. This study was approved by the Ethical Committee of Tongji Hospital, Tongji Medical College, Huazhong University of Science and Technology, Wuhan, China (Grant Number: TJ-C20200128, Date: February 12, 2020).

2.2 | Grouping criteria

The COVID-19 patients were classified into moderate, severe, and critical groups according to "Pneumonia diagnosis and treatment program for novel coronavirus infection (trial version 7)" issued by the National Health Commission of the People's Republic of China.¹⁴ The moderate cases are those with fever, typical symptoms, and pneumonia on chest radiography. Severe cases need to meet one of the following criteria: (1) respiratory distress (respiration rate ≥ 30 times/min); (2) blood oxygen saturation (SpO₂) $\leq 93\%$ in resting state; and (3) arterial partial pressure of O₂ to a fraction of inspired oxygen (PaO₂/FiO₂)

ratio ≤ 300 mmHg. Critical cases meet one of the following criteria: (1) respiratory failure requiring mechanical ventilation; (2) shock; and (3) multiple organ dysfunction needing intensive care unit treatment.

2.3 | SARS-CoV-2 antibody detection

The IgM and IgG antibodies against SARS-CoV-2 in serum specimens were detected using YHLO-CLIA-IgG and YHLO-CLIA-IgM Kits supplied by YHLO (YHLO Biotech Co., Ltd.), according to the manufacturer's instructions. The recombinant antigens contain nucleoprotein and spike (S) protein of SARS-CoV-2. The antibody levels were expressed as arbitrary units per milliliter.

2.4 | Flow cytometry analysis

The percentages and absolute counts of total T cells, CD4⁺ T cells, CD8⁺ T cells, B cells, and NK cells were determined by using Trucount tubes and BD Multitest 6-color TBNK Reagent Kit (BD Biosciences) according to the manufacturer's instructions. In brief, 50 μ l of whole blood was labeled with a 6-color TBNK antibody cocktail for 15 min at room temperature. After adding 450 μ l of FACS Lysing Solution, samples were analyzed with the FACSCanto flow cytometer using FACSCanto clinical software (BD Biosciences).

2.5 | Statistical analysis

Statistical analysis was performed using SPSS 24.0 software and GraphPad Prism 8.3 software. Measurement data were tested for normality. Data that confirmed normality were expressed as mean $\pm$ standard deviation (SD), and t-test was used for comparison between groups. Median and interquartile range (IQR) were used for noncompliant data. The comparison between multiple groups, Kruskal-Wallis rank-sum test and Mann-Whitney U test, was used for pairwise comparison between groups. Pearson correlation tests were also performed. A two-sided α value of less than 0.05 was considered statistically significant.

3 | RESULTS

3.1 | Demographic and clinical characteristics of 707 COVID-19 cases

This study included a total of 707 confirmed COVID-19 patients; among them, 343 (48.5%) were males and 364 (51.5%) were females (Table 1). The cases were classified into three groups, moderate (410 cases, 58.0%), severe (206 cases, 29.1%), and critical (91 cases, 12.9%). The median ages in the moderate group were significantly younger than the severe and critical groups. Most of the patients had fever (67.0%), cough (62.5%), expectoration (18.7%), shortness of

TABLE 1 Baseline characteristics of 707 patients of COVID-19

	Total (n = 707)	Moderate (n = 410)	Severe (n = 206)	Critical (n = 91)
Age, median (IQR) (years)	62.0 (50, 70)	57 (43, 65)	65.5 (58, 75)	67 (57, 73)
Sex				
Male	343 (48.5%)	181 (44.2%)	103 (50.0%)	59 (64.8%)
Female	364 (51.5%)	229 (55.7%)	103 (50.0%)	32 (35.2%)
Signs and symptoms at admission				
Fever	474 (67.0%)	265 (64.6%)	148 (71.8%)	61 (67.0%)
Cough	442 (62.5%)	235 (57.3%)	155 (75.2%)	52 (57.1%)
Expectoration	132 (18.7%)	62 (15.1%)	53 (25.7%)	17 (18.7%)
Shortness of breath	118 (16.7%)	26 (6.3%)	52 (25.2%)	41 (45.1%)
Chest distress or chest pain	93 (12.7%)	56 (13.7%)	24 (11.7%)	3 (3.3%)
Diarrhea	95 (13.4%)	49 (12.0%)	40 (19.4%)	6 (6.6%)
Muscle ache	25 (3.5%)	14 (3.4%)	6 (2.9%)	5 (5.5%)
Pharyngalgia	18 (2.5%)	14 (3.4%)	2 (1.0%)	2 (2.2%)
Nausea and vomiting	12 (1.7%)	2 (0.5%)	4 (1.9%)	6 (6.6%)
Comorbidities				
Hypertension	268 (37.9%)	124 (30.2%)	95 (46.1%)	49 (53.8%)
Diabetes	135 (19.1%)	66 (16.1%)	44 (21.4%)	25 (27.5%)
Cardiovascular disease	79 (11.1%)	35 (8.5%)	30 (14.6%)	14 (15.4%)
Malignancy	31 (4.4%)	14 (3.4%)	7 (3.4%)	10 (11.0%)
Chronic kidney disease	29 (3.8%)	11 (2.7%)	9 (4.4%)	9 (9.9%)
Chronic liver disease	27 (3.8%)	15 (3.7%)	6 (2.9%)	6 (6.6%)
Thyroid disease	26 (3.7%)	15 (3.7%)	6 (2.9%)	5 (5.5%)
COPD	24 (3.4%)	10 (2.4%)	3 (1.5%)	11 (12.1%)
Outcome time, median (IQR) (days)	51 (40, 60)	49 (37, 58)	52 (43, 60)	59 (44, 78)
Clinical outcome				
Recovered	677 (95.8%)	410 (100%)	206 (100%)	61 (67.0%)
Deceased	30 (4.2%)	0 (0%)	0 (0%)	30 (33.0%)

Abbreviations: COPD, chronic obstructive pulmonary disease; COVID-19: coronavirus disease 2019; IQR, interquartile range.

breath (16.7%), chest distress or chest pain (12.7%), and diarrhea (13.4%) at the illness onset. The most common comorbidities of the cases were hypertension (37.9%), diabetes (17.1%), and cardiovascular disease (11.1%). The median days from illness onset to hospital discharge or death were 51 (IQR, 40–60) days, and the median days for the critical group were 59 (IQR, 44–78) days, which were significantly longer than moderate and severe groups. A total of 677 (95.8%) patients of the included cases were discharged and 30 (4.2%) patients were deceased.

3.2 | Laboratory results in COVID-19 patients with different illness severity

The levels of antibody and the counts of lymphocyte subsets in COVID-19 patients with different illness severity were shown in Figure 1. The counts of total T cells, CD4⁺ T cells, CD8⁺ T cells, NK cells, and B cells were significantly decreased with the increased severity of illness, but no statistical difference was found for CD4/CD8 ratio (Figure 1A). The levels of IgM and IgG were higher in

severe cases than moderate cases but the median days from illness onset to detection have no statistical difference. However, the antibody levels in critical cases were significantly lower than moderate and severe cases while with longer median days from illness onset to detection (Figure 1B,C). The laboratory results of critical patients with different clinical outcomes were further compared. The counts of total T cells, CD4⁺ T cells, CD8⁺ T cells, NK cells, and B cells in the deceased group were all significantly lower than the recovered group (Figure 2A), but no significant difference of IgM and IgG levels were observed (Figure 2B). On the contrary, the CD4/CD8 ratio in the deceased cases was higher than the recovered cases (Figure 2A).

3.3 | The relationship between the levels of antibodies against SARS-CoV-2 and the amounts of lymphocyte subsets

The levels of IgM and IgG antibodies and the counts of lymphocyte subsets in one patient were assessed during the same day. Our data

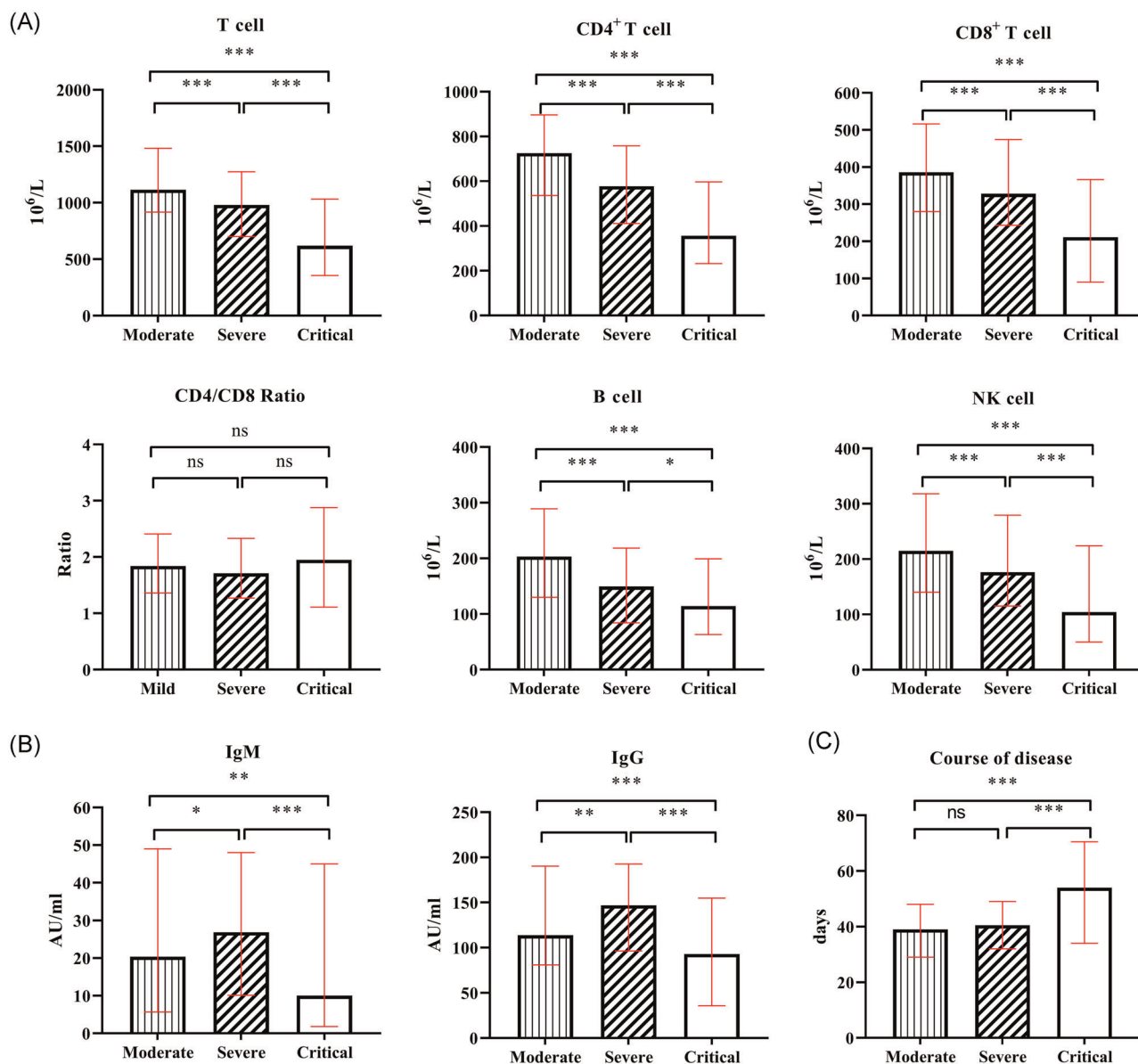


FIGURE 1 The levels of antibody and the amounts of lymphocyte subsets in COVID-19 patients with different severity of illness. A, The median levels of lymphocyte subsets (total T cells, CD4⁺ T cells, CD8⁺ T cells, CD4/CD8 ratio, B cells, and NK cells) in different groups were shown. B, The median levels of IgM and IgG in different groups were shown. C, The median days from illness onset to antibody detection were shown. Data were presented as the median (interquartile range) and were analyzed using the Mann-Whitney test. All statistical analyses were performed using GraphPad Prism 8.3 (* $p < .05$, ** $p < .01$, *** $p < .001$, ns $p > .05$). COVID-19, coronavirus disease 2019; IgG, immunoglobulin G; NK, natural killer. $p < .05$ was considered statistically significant

show that no association was observed between antibody levels and the T cells, CD4⁺ T cells, CD8⁺ T cells, NK cells, and B cells (Table 2).

3.4 | The dynamic changes of antibody levels and the counts of lymphocyte subsets in COVID-19 patients with different illness severity

The dynamic changes of antibody levels and lymphocyte subsets of 707 COVID-19 patients with different illness severity were shown in Figure 3. The counts of lymphocyte subsets in severe and

critical illness patients were lower than those in moderate patients during the course of disease. There were no significant changes in NK cells and B cells among the three groups. The total T cells, CD4⁺ T cell, and CD8⁺ T cell in moderate patients showed no obvious changes during the course of disease. In the severe group and critical group, the counts of lymphocyte subsets were lower than the moderate group at disease onset but gradually recovered to the normal and showed stable levels around 43 days after illness onset. The index of CD4/CD8 ratio in critical cases gradually decreased to the levels of moderate and severe cases around 33 days from illness onset (Figure 3A). For the antibody response

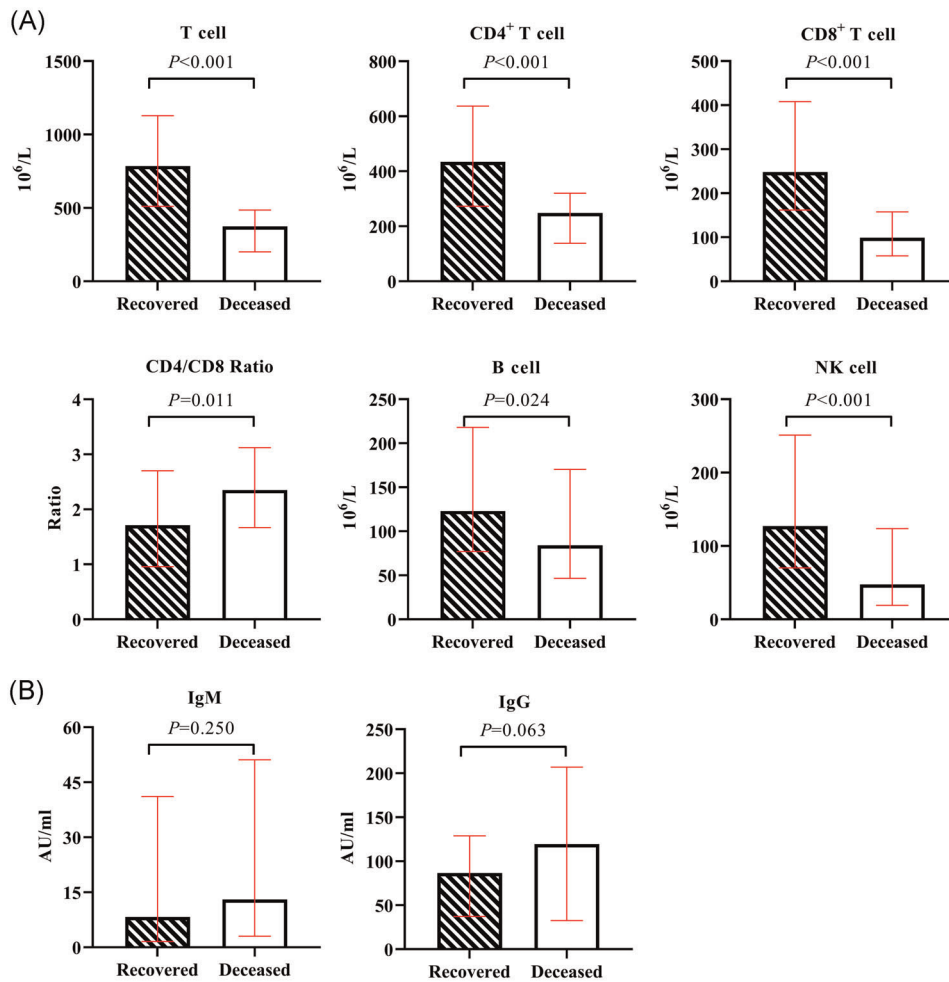


FIGURE 2 Laboratory results of critical COVID-19 patients with different clinical outcomes. A, The median levels of lymphocyte subsets (total T cells, CD4⁺ T cells, CD8⁺ T cells, CD4/CD8 ratio, B cells, and NK cells) in recovered and deceased cases were shown. B, The median levels of IgM and IgG in recovered and deceased cases were shown. Data were presented as the median (interquartile range) and were analyzed using the Mann-Whitney test. COVID-19, coronavirus disease 2019; IgG, immunoglobulin G; NK, natural killer. $p < .05$ was considered statistically significant

after SARS-CoV-2 infection, IgM began to increase within 1–2 weeks after illness onset in moderate patients, reached a stable level in 25 days, and then began to decline in 43 days. The increase of IgM was delayed in severe and critical groups than in the moderate group. For the response of IgG antibody, patients in

moderate and severe could increase in the early stage of infection, and reached a peak level after 2 weeks, while in the critical patients the production of IgG antibody was later than in other moderate and severe patients, but both can maintain a stable level in 48 days after illness onset (Figure 3B).

TABLE 2 The relationship between the levels of antibody and the amounts of lymphocyte subsets in COVID-19 patients

	Total lymphocytes	T cell	CD4 ⁺ T cell	CD8 ⁺ T cell	NK cell	B cell
IgG	$r = .03$ $p = .351$	$r = .029$ $p = .370$	$r = .044$ $p = .163$	$r = -.013$ $p = .690$	$r = -.009$ $p = .773$	$r = .046$ $p = .150$
IgM	$r = .002$ $p = .939$	$r = .009$ $p = .077$	$r = .012$ $p = .718$	$r = -.007$ $p = .825$	$r = -.018$ $p = .580$	$r = .001$ $p = .965$

Note: $r > 0$ represents two variables positive correlation; $r < 0$ represents two variables negative correlation. $|r| \geq .8$, high correlated; $.5 \leq |r| < .8$, moderately correlated; $.3 \leq |r| < .5$, low correlation; $|r| \leq .3$, weak and basically irrelevant. $p < .05$ was considered statistically significant.

Abbreviations: COVID-19, coronavirus disease 2019; IgG, immunoglobulin G; NK, natural killer.

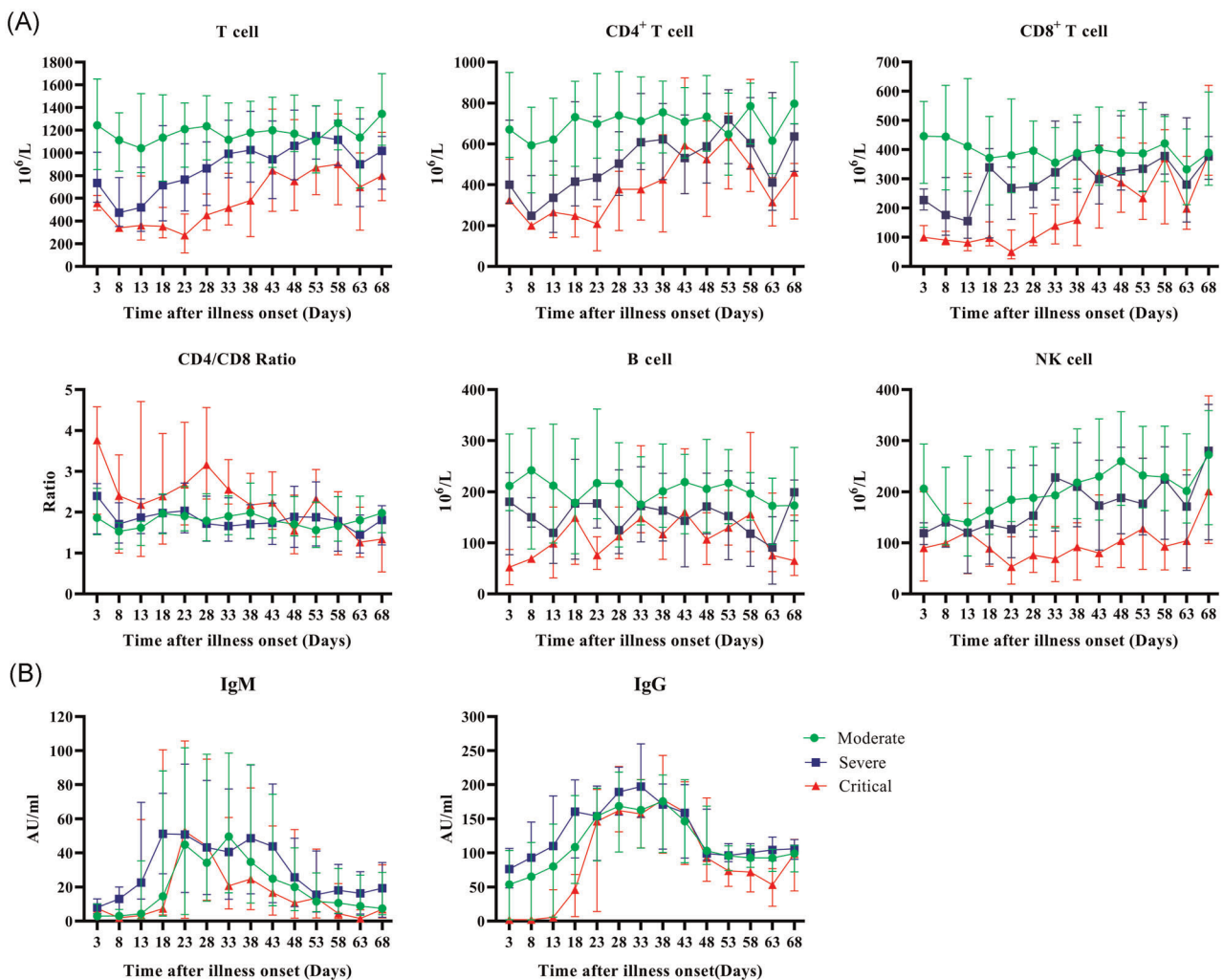


FIGURE 3 The dynamic change of antibody and lymphocyte subsets in COVID-19 patients with different illness severity. A, The dynamic change of lymphocyte subsets (total T cells, CD4⁺ T cells, CD8⁺ T cells, CD4/CD8 ratio, B cells, and NK cells) in COVID-19 patients with different severity of illness were shown. B, The dynamic changes of antibody levels (IgM and IgG) in COVID-19 patients with different severity of illness were shown. Data are presented as the median (interquartile range). COVID-19, coronavirus disease 2019; IgG, immunoglobulin G; NK, natural killer

4 | DISCUSSION

COVID-19 has spread worldwide and become an urgent public health challenge.¹⁵ At present, about half of the infected patients have severe pneumonia, nearly one-third of patients have acute respiratory distress syndrome (ARDS), and once the infected patient progresses to ARDS, more than 10% of patients in a short time internal deterioration, died of multiple organ failure.^{16–19} Because there is no therapeutic vaccine or specific antiviral drugs so far, monitoring the dynamic changes of the immune status of COVID-19 patients is essential for early detection and intervention of potentially serious patients. Previous studies have reported the characteristics of various laboratory markers in patients with COVID-19,^{6,20} but the dynamic changes of specific antibodies and lymphocyte subsets in patients with different illness severity still need further investigation.

In this study, we described 707 COVID-19 patients with different illness severity. Most patients had fever, cough, sputum, and shortness of breath. The comorbidities are mainly hypertension, diabetes, and cardiovascular diseases. The median age and duration of disease in severe and critically ill patients are higher than those in moderate patients, which are basically consistent with previous research reports.^{4,17} Previous studies have shown that cell-mediated and humoral immune responses play a vital role in the defense and clearance of SARS-CoV-2.^{6,20–22} To assess the host immunity response comprehensively, lymphocyte counts and antibody levels should be done simultaneously. Our data show that no correlation was observed between antibody levels and the counts of lymphocyte subsets in COVID-19 patients, which might be due to that the detection of lymphocyte subsets could not reflect the specific T cell or plasma cell levels during SARS-CoV-2 infection. In addition, the level of IgG antibody against the receptor-binding domain of S protein has

been shown to be protective against SARS-CoV-2 infection and correlated with the titer of neutralizing antibody and the specific T cells.²³ Therefore, specific T cell and plasma cell levels need to be further investigated to evaluate the immune responses more accurately in COVID-19 patients.

Lymphopenia has been used as one useful indicator for the diagnosis, predicting the progression and prognosis of COVID-19 patients.^{24,25} Our data also show that the counts of total T cells, CD4⁺ T cells, CD8⁺ T cells, B cells, and NK cell counts were significantly decreased with the increase of illness severity, and a further decrease of lymphocyte subsets was observed in the deceased cases, which further validates the important value of lymphocyte subsets for predicting the prognosis of COVID-19. Previous studies have highlighted the importance of T cells, especially CD4⁺ T cells, rather than B cells, in controlling and improving the pathogenesis of SARS-CoV and MERS-CoV infections.²⁶ By analyzing the changes of lymphocyte subsets during the course of disease, the levels of T cells, CD4⁺ T cells, and CD8⁺ T cells in severe and critical cases were lower than those in moderate cases, but no significant changes of NK cells and B cells were observed in patients with different illness severity. Some studies have shown that the increased CD4/CD8 ratio is an independent risk factor for poor efficacy in COVID-19 patients.⁸ Our data show that the CD4/CD8 ratio in critical patients was higher than that of moderate and severe patients and gradually decreased to the normal level during the course of disease, which was consistent with the reported studies. Therefore, longitudinal detection of the lymphocyte subsets might provide information for predicting the progression and outcomes of COVID-19 patients.

Rapid and specific IgM and IgG antibody detection could offer information for confirmation or exclusion of SARS-CoV-2 infection in suspected patients. We found that the levels of antibodies were higher in severe patients than moderate and critical patients. The levels of IgM and IgG in critical illness patients are the lowest in the present study, which may be due to the longer virus exposure or severely impaired immune response in these patients. No difference was found of antibody levels between deceased patients and recovered patients. Some studies have found that the antibody levels were higher in the severe groups than the nonsevere groups at any stage of the disease.²⁷ But other studies concluded that antibody levels could not be used to predict the severity of the disease.^{28,29} The kinetic changes of antibody levels suggested that the production of IgM and IgG was delayed in critical patients than moderate and severe patients. The possible explanation might be the longer days from illness onset and the antibodies gradually decreased to the near background levels or the impaired immune response. High levels of IgG were observed in the early stage of moderate and severe patients, as reported in other studies.^{29,30} So quantitative IgM and IgG assays might indicate the prognosis of COVID-19.

However, our study has certain limitations. First, it was a retrospective, single-center study of patients admitted to the hospital; standardized data for a larger cohort would be better to assess the temporal change of immune response after infection with COVID-19. Second, the time from symptom onset to admission may be long and the continuous monitoring data in one patient were limited. Third,

we only analyzed peripheral blood lymphocytes. Further analysis of lymphocytes from alveolar lavage fluid is needed.

In conclusion, this study found that T cells, CD4⁺ T cells, CD8⁺ T cells, and CD4/CD8 ratio are helpful markers to indicate the severe illness of COVID-19. The dynamic changes of lymphocyte subsets and antibodies differ among COVID-19 patients with different illness severity, which have potential value in understanding the pathogenesis of COVID-19.

ACKNOWLEDGMENTS

This study was funded by the National Mega Project on Major Infectious Disease Prevention (Grant No. 2017ZX10103005-007).

CONFLICT OF INTERESTS

The authors declare that there are no conflict of interests.

AUTHOR CONTRIBUTIONS

Bo Zhang and Hongyan Hou designed the study, Daoyuan Yue and Yun Wang collected the data, Bo Zhang and Feng Wang analyzed the data, Bo Zhang and Hongyan Hou write the paper, Shiji Wu reviewed the paper.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

ORCID

Shiji Wu  <https://orcid.org/0000-0002-0238-6926>

REFERENCES

- Jiang S, Shi Z, Shu Y, et al. A distinct name is needed for the new coronavirus. *Lancet*. 2020;395(10228):949. [https://doi.org/10.1016/S0140-6736\(20\)30419-0](https://doi.org/10.1016/S0140-6736(20)30419-0)
- Vellingiri B, Jayaramayya K, Iyer M, et al. COVID-19: a promising cure for the global panic. *Sci Total Environ*. 2020;725:138277. <https://doi.org/10.1016/j.scitotenv.2020.138277>
- Coronaviridae Study Group of the International Committee on Taxonomy of Viruses. The species severe acute respiratory syndrome-related coronavirus: classifying 2019-nCoV and naming it SARS-CoV-2. *Nat Microbiol*. 2020;5(4):536-544. <https://doi.org/10.1038/s41564-020-0695-z>
- Wang D, Hu B, Hu C, et al. Clinical characteristics of 138 hospitalized patients with 2019 novel coronavirus-infected pneumonia in Wuhan, China. *J Am Med Assoc*. 2020;323(11):1061-1069. <https://doi.org/10.1001/jama.2020.1585>
- Lai CC, Shih TP, Ko WC, Tang HJ, Hsueh PR. Severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) and coronavirus disease-2019 (COVID-19): the epidemic and the challenges. *Int J Antimicrob Agents*. 2020;55(3):105924. <https://doi.org/10.1016/j.ijantimicag.2020.105924>
- Li G, Fan Y, Lai Y, et al. Coronavirus infections and immune responses. *J Med Virol*. 2020;92(4):424-432. <https://doi.org/10.1002/jmv.25685>
- Cecere TE, Todd SM, Leroith T. Regulatory T cells in arterivirus and coronavirus infections: do they protect against disease or enhance it? *Viruses*. 2012;4(5):833-846. <https://doi.org/10.3390/v4050833>
- Wang F, Nie J, Wang H, et al. Characteristics of peripheral lymphocyte subset alteration in COVID-19 pneumonia. *J Infect Dis*. 2020;221(11):1762-1769. <https://doi.org/10.1093/infdis/jiaa150>

9. Du RH, Liang LR, Yang CQ, et al. Predictors of mortality for patients with COVID-19 pneumonia caused by SARS-CoV-2: a prospective cohort study. *Eur Respir J*. 2020;55(5):2000524. <https://doi.org/10.1183/13993003.00524-2020>
10. Wang F, Hou H, Luo Y, et al. The laboratory tests and host immunity of COVID-19 patients with different severity of illness. *JCI Insight*. 2020;5(10):137799. <https://doi.org/10.1172/jci.insight.137799>
11. Racine R, Winslow GM. IgM in microbial infections: taken for granted? *Immunol Lett*. 2009;125(2):79-85. <https://doi.org/10.1016/j.imlet.2009.06.003>
12. Hou H, Wang T, Zhang B, et al. Detection of IgM and IgG antibodies in patients with coronavirus disease 2019. *Clin Transl Immunol*. 2020;9(5):e01136. <https://doi.org/10.1002/cti2.1136>
13. Jin Y, Wang M, Zuo Z, et al. Diagnostic value and dynamic variance of serum antibody in coronavirus disease 2019. *Int J Infect Dis*. 2020;94:49-52. <https://doi.org/10.1016/j.ijid.2020.03.065>
14. National Health Commission: Diagnosis and Treatment of Pneumonia Infected by Novel Coronavirus (trial 7th ed.) China. <http://www.nhc.gov.cn/yzygj/s7653p/03/46c9294a7dfe4cef80dc7f5912eb1989/files/ce3e6945832a438eaae415350a8ce964.pdf>. Accessed March 03, 2020.
15. Kucharski AJ, Russell TW, Diamond C, et al. Early dynamics of transmission and control of COVID-19: a mathematical modelling study. *Lancet Infect Dis*. 2020;20(5):553-558. [https://doi.org/10.1016/S1473-3099\(20\)30144-4](https://doi.org/10.1016/S1473-3099(20)30144-4)
16. Chen N, Zhou M, Dong X, et al. Epidemiological and clinical characteristics of 99 cases of 2019 novel coronavirus pneumonia in Wuhan, China: a descriptive study. *Lancet*. 2020;395(10223):507-513. [https://doi.org/10.1016/S0140-6736\(20\)30211-7](https://doi.org/10.1016/S0140-6736(20)30211-7)
17. Huang C, Wang Y, Li X, et al. Clinical features of patients infected with 2019 novel coronavirus in Wuhan, China. *Lancet*. 2020;395(10223):497-506. [https://doi.org/10.1016/S0140-6736\(20\)30183-5](https://doi.org/10.1016/S0140-6736(20)30183-5)
18. Liu Y, Yang Y, Zhang C, et al. Clinical and biochemical indexes from 2019-nCoV infected patients linked to viral loads and lung injury. *Sci China Life Sci*. 2020;63(3):364-374. <https://doi.org/10.1007/s11427-020-1643-8>
19. Zhang J, Dong X, Cao Y, et al. Clinical characteristics of 140 patients infected with SARS-CoV-2 in Wuhan, China. *Allergy*. 2020;75:1730-1741. <https://doi.org/10.1111/all.14238>
20. Rokni M, Ghasemi V, Tavakoli Z. Immune responses and pathogenesis of SARS-CoV-2 during an outbreak in Iran: comparison with SARS and MERS. *Rev Med Virol*. 2020;30(3):e2107. <https://doi.org/10.1002/rmv.2107>
21. Baruah V, Bose S. Immunoinformatics-aided identification of T cell and B cell epitopes in the surface glycoprotein of 2019-nCoV. *J Med Virol*. 2020;92(5):495-500. <https://doi.org/10.1002/jmv.25698>
22. Azkur AK, Akdis M, Azkur D, et al. Immune response to SARS-CoV-2 and mechanisms of immunopathological changes in COVID-19. *Allergy*. 2020;75(7):1564-1581. <https://doi.org/10.1111/all.14364>
23. Ni L, Ye F, Cheng ML, et al. Detection of SARS-CoV-2-specific humoral and cellular immunity in COVID-19 convalescent individuals. *Immunity*. 2020;52(6):971-977.e3. <https://doi.org/10.1016/j.immuni.2020.04.023>
24. COVID-19 Diagnosis and Treatment Guideline in China (7th ed.). National Health Commission of the People's Republic of China. <http://www.nhc.gov.cn/yzygj/s7653p/03/46c9294a7dfe4cef80dc7f5912eb1989.shtml>. Accessed March 17, 2020.
25. Yan L, Zhang HT, Goncalves J, et al. An interpretable mortality prediction model for COVID-19 patients. *Nat Mach Intell*. 2020;2:283-288. <https://doi.org/10.1038/s42256-020-0180-7>
26. Zhao J, Li K, Wohlford-Lenane C, et al. Rapid generation of a mouse model for Middle East respiratory syndrome. *Proc Natl Acad Sci USA*. 2014;111(13):4970-4975. <https://doi.org/10.1073/pnas.1323279111>
27. Ou J, Tan M, He H, et al. Study on the expression levels of antibodies against SARS-CoV-2 at different period of disease and its related factors in 192 cases of COVID-19 patients. *medRxiv*. 2020:2020-2025. <https://doi.org/10.1101/2020.05.22.20102525>
28. Phipps WS, Sorelle JA, Li Q, et al. SARS-CoV-2 Antibody responses do not predict COVID-19 disease severity. *medRxiv*. 2020:2020-2025. <https://doi.org/10.1101/2020.05.15.20103580>
29. Long QX, Liu BZ, Deng HJ, et al. Antibody responses to SARS-CoV-2 in patients with COVID-19. *Nat Med*. 2020;26:845-848. <https://doi.org/10.1038/s41591-020-0897-1>
30. To KKW, Tsang OTY, Leung WS, et al. Temporal profiles of viral load in posterior oropharyngeal saliva samples and serum antibody responses during infection by SARS-CoV-2: an observational cohort study. *Lancet Infect Dis*. 2020;20(5):565-574. [https://doi.org/10.1016/S1473-3099\(20\)30196-1](https://doi.org/10.1016/S1473-3099(20)30196-1)

How to cite this article: Zhang B, Yue D, Wang Y, Wang F, Wu S, Hou H. The dynamics of immune response in COVID-19 patients with different illness severity. *J Med Virol*. 2021;93:1070-1077. <https://doi.org/10.1002/jmv.26504>